

GAOUA, Burkina Faso

WMO: 655220

Lat: 10.333N

Lon: 3.183W

Elev: 335

StdP: 97.36

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.3	16.4	-7.6	2.1	31.2	-5.5	2.5	30.9	6.5	30.7	6.3	30.3	0.1	20	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	12.0	38.4	20.8	37.6	20.9	36.9	21.0	26.6	31.9	26.2	31.6	25.8	31.3	2.3	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.4	21.4	28.5	24.9	20.8	28.3	24.5	20.4	28.0	85.1	31.7	83.4	31.6	81.8	31.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.7	5.1	DB	12.6	40.1	1.6	0.6	11.4	40.5	10.4	40.9	9.5	41.2	8.3	41.7	
			WB	7.8	27.9	0.8	0.6	7.2	28.3	6.8	28.7	6.3	29.1	5.7	29.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	25.4	28.3	31.1	31.1	29.7	28.0	26.6	26.0	26.6	27.7	27.8	25.6
	DBStd	2.44	1.60	2.25	1.71	1.82	1.72	1.60	1.29	1.09	1.26	1.23	1.27	1.49
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6502	477	513	653	633	611	541	515	497	497	550	533	483
	CDD18.3	3461	219	280	394	383	352	291	257	238	247	292	283	224
	CDH23.3	40806	2779	3791	5702	5444	4576	3279	2337	1887	2080	2910	3264	2757
CDH26.7	20728	1517	2222	3467	3191	2362	1352	754	506	708	1309	1818	1522	
Wind	WSAvg	1.8	1.6	1.7	2.0	2.5	2.4	2.3	2.1	1.8	1.2	1.2	1.1	1.5
Precipitation	PrecAvg	1066	3	5	28	73	118	133	186	228	185	86	11	5
	PrecMax	1564	95	58	103	222	245	262	412	580	356	311	77	74
	PrecMin	713	0	0	0	6	37	40	46	75	92	1	0	0
	PrecStd	168	14	12	25	43	45	48	65	89	59	57	17	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.9	38.9	39.3	39.1	37.5	35.6	33.1	32.1	33.5	35.6	36.9	36.5
		MCWB	17.2	19.4	19.8	22.4	24.8	25.3	25.2	25.2	25.6	23.0	19.8	17.7
	2%	DB	35.7	37.8	38.6	38.0	36.3	34.2	32.0	31.0	32.3	34.4	36.1	35.5
		MCWB	16.9	19.2	20.2	23.0	25.0	25.2	24.9	24.8	25.4	24.1	20.2	17.4
	5%	DB	34.6	36.9	37.8	37.1	35.2	33.2	31.2	30.2	31.4	33.5	35.4	34.6
		MCWB	16.5	18.8	20.4	23.4	25.1	25.1	24.7	24.6	25.2	24.6	20.6	17.2
	10%	DB	33.5	35.9	37.0	36.1	34.2	32.1	30.3	29.3	30.4	32.6	34.5	33.6
		MCWB	15.9	18.1	20.6	23.7	25.1	24.9	24.6	24.5	25.0	24.8	21.1	17.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	24.7	25.7	26.2	27.1	26.6	26.0	26.1	26.7	27.0	25.4	24.0
		MCDB	30.1	32.0	33.1	33.8	33.0	31.5	30.3	29.8	31.0	31.0	31.2	31.2
	2%	WB	20.2	23.9	25.0	25.8	26.6	26.0	25.5	25.5	26.1	26.2	24.8	22.6
		MCDB	30.3	32.2	32.9	33.2	32.7	31.6	29.9	29.3	30.4	31.0	30.8	28.6
	5%	WB	17.9	23.1	24.5	25.4	26.1	25.6	25.1	25.1	25.6	25.7	24.3	20.0
		MCDB	32.2	31.3	32.5	32.9	32.3	31.0	29.6	28.8	29.8	30.7	30.4	30.0
	10%	WB	16.7	22.0	24.0	25.0	25.7	25.2	24.8	24.7	25.3	25.3	23.8	18.4
		MCDB	31.4	30.8	32.0	32.7	31.8	30.4	29.1	28.2	29.2	30.2	29.8	30.3
Mean Daily Temperature Range	5% DB	MDBR	16.8	14.9	12.0	9.9	9.1	7.8	7.2	6.6	7.7	10.2	15.1	17.4
		MCDBR	18.5	16.0	13.4	11.1	10.2	9.2	8.5	7.9	9.3	11.7	16.2	18.5
		MCWBR	5.8	5.2	4.5	3.2	2.8	2.6	2.8	2.9	3.4	3.5	4.2	5.1
	5% WB	MCDBR	17.2	11.8	10.3	9.8	9.4	8.2	7.9	7.3	8.7	10.3	12.5	16.0
		MCWBR	6.4	5.2	3.8	3.3	3.4	3.0	2.9	3.1	3.6	3.9	3.6	6.1
Clear-Sky Solar Irradiance	taub	0.598	0.670	0.772	0.675	0.602	0.555	0.503	0.481	0.494	0.557	0.558	0.550	
	taud	1.767	1.647	1.511	1.739	1.852	1.978	2.172	2.247	2.180	1.970	1.897	1.888	
	Ebn at Noon	722	688	630	690	727	751	796	826	824	767	750	751	
	Edh at Noon	223	259	302	238	208	181	150	141	152	185	194	194	
All-Sky Solar Radiation	RadAvg	5.55	5.71	5.87	5.96	5.84	5.34	4.94	4.61	5.00	5.63	5.48	5.37	
	RadStd	0.27	0.26	0.22	0.19	0.23	0.22	0.20	0.19	0.19	0.23	0.15	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page